

Chest Xray Disease Classification Using Deep Learning

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ABSTRACT Chest X-ray (CXR) imaging is routinely used to investigate lung and cardiac diseases, yet reliable reporting still depends on experienced radiologists, who may not be available in all settings. This paper reviews how deep learning techniques from artificial intelligence can be applied to automate CXR interpretation for disease detection. It introduces the basic principles of deep neural networks for X-ray analysis and explains how strategies such as transfer learning and data augmentation are used to improve performance. Recent work on detecting lung nodules, pneumonia, pneumothorax and other common abnormalities is summarized, with particular attention to models that predict multiple pathologies simultaneously. The review compares model performance with human readers and discusses practical barriers to using these systems in everyday clinical workflows, including how they integrate with radiologists reporting. Overall, the paper evaluates whether deep learning can improve the accuracy and accessibility of chest X-ray diagnosis, especially in environments with limited specialized expertise.

INDEX TERMS Chest X-rays, Deep Learning, Disease Detection, Artificial Intelligence, Neural Networks.

I. INTRODUCTION

Chest X-rays (CXRs) remain one of the most frequently requested tests for assessing thoracic and cardiac conditions in routine practice. They are attractive because they are inexpensive, quick to acquire and provide a broad overview of chest anatomy in a single exposure. However, ribs, mediastinum and lung fields overlap on the projection image, and contrast between tissues can be subtle, so small nodules or limited areas of consolidation may be overlooked. At the same time, demand for CXRs has increased more rapidly than the number of radiologists able to read them, adding pressure to reporting workflows. Against this background, artificial-intelligence (AI) tools are being explored to assist with CXR interpretation by detecting, segmenting and describing anatomical structures and abnormalities, with early evaluations suggesting benefits for both accuracy and efficiency.

A. TECHNICAL BACKGROUND

In medical imaging, most AI systems are now built on machine-learning methods that learn patterns directly from data instead of relying on fixed, hand-coded rules. Early work used classical models such as support vector machines and decision trees, but deep neural networks (DNNs) have become dominant because their stacked layers can capture increasingly abstract image features. This differs from traditional computer-aided detection pipelines, where feature extraction and classification were designed separately by experts. In DNNs the same network learns both steps jointly, although this usually requires large, well-annotated datasets. Convolutional neural networks (CNNs) are the main DNN architecture for image data; they apply small learnable filters across the image, combine multiple convolution and pooling layers, and produce high-level representations that can be used for classification or segmentation tasks on CXRs.

Because big labelled medical datasets are expensive to create, many studies use transfer learning: they start from CNNs pre-trained on large natural-image datasets such as ImageNet and then fine-tune them on smaller, domain-specific collections like chest X-rays. This strategy often gives good performance with less data and speeds up models development for clinical applications. Finally, visualization tools help make these models more interpretable. Class activation maps and related methods overlay heatmaps on the X-ray to show which regions contributed most to a decision, giving radiologists a visual explanation of the model's focus when suggesting a diagnosis.

II. DATASET

Large, well-labelled datasets are a key requirement for training and testing AI systems in medical imaging. For chest X-ray (CXR) work, researchers rely heavily on public resources such as ChestX-ray8 and ChestX-ray14, which include over one hundred thousand images with labels for common findings like pneumonia, pneumothorax, nodules, and masses. Beyond sheer size, the type and quality of labels matter. Many datasets use weak labels, where each image only has image-level tags (for example “pneumonia” or “no finding”) and no precise indication of where the abnormality lies, which limits how finely a model can localise disease. Strong labels provide detailed region-level or pixel-level annotations that are much more informative for training but are slow and expensive for experts to create. These datasets are also highly imbalanced: some diseases are very frequent while others have relatively few examples. This imbalance can bias models towards the common classes, so studies often apply data-augmentation and related strategies to synthetically increase minority-class examples and improve robustness

III. MEASURING ML PERFORMANCE

Evaluating AI-assisted diagnostic systems for chest X-ray interpretation requires rigorous performance metrics that capture both sensitivity and specificity across a range of disease patterns. Sensitivity and specificity, which measure how well a model detects true positives and rejects true negatives, form the basic indicators of diagnostic accuracy. The area under the ROC curve (AUROC) condenses this trade-off into a single number by measuring how well the model separates normal from abnormal cases across all possible decision thresholds. For tasks that require localising lesions, free-response ROC (FROC) curves plot the proportion of correctly localised findings against the average number of false-positive marks per image, giving a more detailed view of detection behaviour. For segmentation tasks, overlap-based measures such as the Dice similarity coefficient quantify how closely the predicted masks match expert annotations, reflecting both segmentation accuracy and boundary quality. By combining metrics that assess image-level classification, lesion-level

detection, and pixel-level segmentation, researchers can obtain a more complete picture of the strengths and limitations of AI systems and use these insights to refine models and support safer deployment in radiology practice.

IV. AUTOMATIC DISEASE DETECTION ON CXR IMAGES

Deep learning (DL) has the potential to substantially change how diagnostic tasks are performed in medicine. Although DL can be applied to many problems, from automatically drafting radiology reports to mining large image archives for research, this review concentrates on its use for automated analysis of chest X-ray (CXR) images. Driven by the clinical importance of CXRs and the push to integrate AI into routine workflows, a wide range of image-analysis tasks in chest radiology are now being developed and tested using DL methods.

A. PNEUMOTHORAX DETECTION

One important domain where deep neural networks (DNNs) have advanced rapidly is the detection, segmentation, and quantification of pneumothorax on chest X-ray images. These methods mark a major step forward compared with earlier CXR tools and can strongly influence patient care, for example by helping clinicians prioritise treatment according to pneumothorax size and follow changes on serial images. Earlier work relied on traditional machine-learning (ML) techniques for pneumothorax detection and reported only moderate accuracies, typically in the range of about 77% to 88%. More recent studies have shifted towards deep-learning (DL) approaches to improve performance, such as the work by Taylor and colleagues, who trained several DNN models on labelled CXRs with moderate-to-large pneumothoraces and achieved high sensitivity and specificity on held-out test sets

B. EDEMA DETECTION

Deep learning (DL) techniques have become central to improving how pulmonary edema is detected on chest X-ray images. Building on earlier machine-learning approaches, research groups have developed deep neural network (DNN) models that can consistently pick up radiographic signs of fluid overload in the lungs. In many of these studies, convolutional neural networks (CNNs) show particularly strong performance in distinguishing edema patterns from normal lung parenchyma on CXRs. When trained on large, well-labelled datasets and carefully tuned, such models achieve high diagnostic accuracy, indicating that they could provide useful support to clinicians managing patients with suspected pulmonary edema.

C. CARDIOMEGALY DETECTION

Cardiomegaly, or enlargement of the heart, is another condition where deep learning (DL) methods have shown strong promise for automatic detection on chest X-ray images. Using DL algorithms, researchers have developed

models that can consistently identify abnormal cardiac silhouettes suggestive of cardiomegaly. These tools combine sophisticated image-processing pipelines with large annotated datasets and can achieve high sensitivity and specificity for recognising cardiac enlargement. DL-based systems can also integrate additional clinical variables and contextual information, which may further refine predictions and support earlier intervention in patients with cardiovascular disease.

D. MASS DETECTION

Deep learning (DL) algorithms have also proved highly effective for automatically detecting masses, including tumours and nodules, on chest X-ray images. Using convolutional neural networks (CNNs) and related deep architectures, researchers have created models that can accurately localise and describe pulmonary masses with high precision. These systems rely on sophisticated feature-extraction pipelines and extensive data-augmentation strategies to boost the sensitivity and specificity of mass detection. By flagging suspicious lesions automatically, DL-based tools can help radiologists read studies more quickly and support earlier intervention in patients with potentially malignant pulmonary abnormalities

V. ALGORITHM

The core of this work is a deep learning model designed to handle the complexity and subtle findings present on chest X-ray images. The following subsection outlines the network architecture, the training strategy, and the reasoning behind the main technical choices. Model architecture: the starting point in this project is a DenseNet-121 network pre-trained on ImageNet, which has been widely used and shown to perform strongly on medical imaging tasks. The original classification head is removed and replaced with a global average pooling layer and a fully connected layer with 14 sigmoid units, one for each thoracic pathology, enabling multi-label predictions for every image. Training procedure: Optimisation is carried out using the Adam optimiser with a learning rate of $1e-4$, which provides stable *updates* for fine-tuning the pretrained DenseNet121 backbone. To address the strong class imbalance in ChestX-ray14, a class-weighted binary cross-entropy loss is used, giving more importance to rare positive findings so that they are not overwhelmed by the large number of normal cases. Early stopping is applied based on validation loss, and batch size and the numbers of training and validation steps per epoch are chosen to match the dataset splits, which helped stabilise training and improve performance compared with initial runs. Data preprocessing: each chest X-ray is resized to 320×320 pixels and normalised on a per-image basis to reduce exposure-related variation and stabilise optimisation. Augmentation techniques: to enhance generalisation across different scanners and clinical environments, standard data-augmentation operations are applied. Each image can be randomly rotated by a small angle, horizontally flipped, shifted and slightly zoomed, which mimics realistic changes in patient positioning and imaging conditions and effectively enlarges the training set without requiring additional labelled data.

VI. EXPERIMENTS

This section describes the dataset, experimental protocol, and evaluation measures used to assess the performance of the deep learning model.

Dataset: The experiments are based on the NIH ChestX-ray14 dataset, a public collection of 112,120 frontal CXRs from 30,805 patients with labels for 14 thoracic conditions. Due to computational constraints, this project uses a reduced training subset (“train-small”) together with separate validation and test splits. While this limits the absolute performance that can be achieved, it still allows us to investigate how transfer learning and data augmentation behave on this task.

Experimental setup: The available images are divided into three disjoint subsets for training, validation and testing, providing separate data for model learning, hyper-parameter tuning and final evaluation. In the final configuration, the data are divided into training, validation, and test sets with approximate proportions of 60%, 20% and 20%, and standard data-augmentation operations (small rotations, horizontal flips, translations and zooms) are applied on the training set to improve generalisation over the earlier baseline model that achieved around 56% test binary accuracy

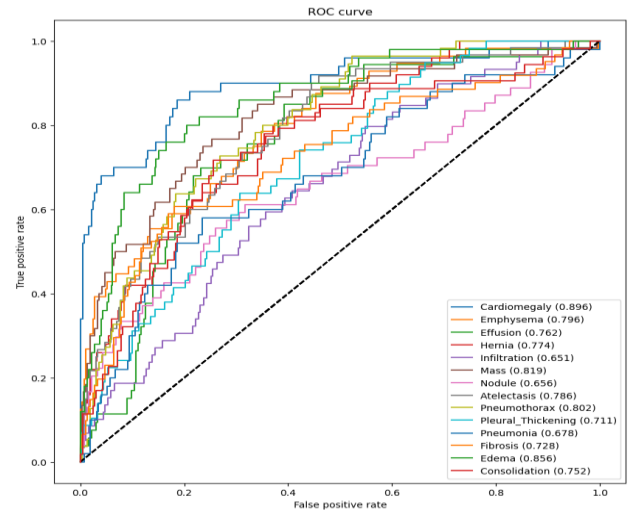


Fig. 1 "The performance of the model in accurately detecting positive instances of each condition." Area Under the Receiver Operating Characteristic Curve (AUC-ROC) : AUC-ROC is a comprehensive metric that assesses how well a model distinguishes between classes across all classification thresholds.

Results analysis: these metrics, together with ROC curves and per-class F1-scores, help highlight which disease classes the model handles relatively well and which remain challenging. They also make it clear that accuracy alone can overestimate performance when many classes are rare, so AUC-ROC and F1-score provide a more nuanced view of the model's behaviour.

Quantitative results: On the held-out test set, the final DenseNet-121 model trained on the train-small subset achieves a test binary accuracy of approximately 76%. When evaluated using more appropriate metrics for this imbalanced, multi-label task, the model reaches a macro AUC of about 0.60 across the 14 pathologies, with some diseases such as Edema and Pneumothorax achieving AUC values around 0.70. Using a fixed decision threshold of 0.5, the macro F1-score is roughly 0.24, and it increases to about 0.29 when a separate threshold is tuned for each class to maximise F1. The improvement in F1 score with class-specific thresholds reflects the fact that a single global cut-off of 0.5 is sub-optimal for this highly imbalanced, multilabel problem. These results show that the model is clearly learning useful signal beyond random guessing, while also highlighting that performance remains modest compared with systems trained on the full dataset with more extensive tuning.

Accuracy improvements: To illustrate the impact of successive optimisation steps, a bar chart was generated comparing four model configurations: an initial baseline, a version with data augmentation and extended training, a configuration with corrected training and validation step counts and early stopping, and the final tuned model. Test binary accuracy increased from roughly 57% in the baseline to 66%, then 73%, and finally around 76% in the best configuration, highlighting the cumulative benefits of these changes

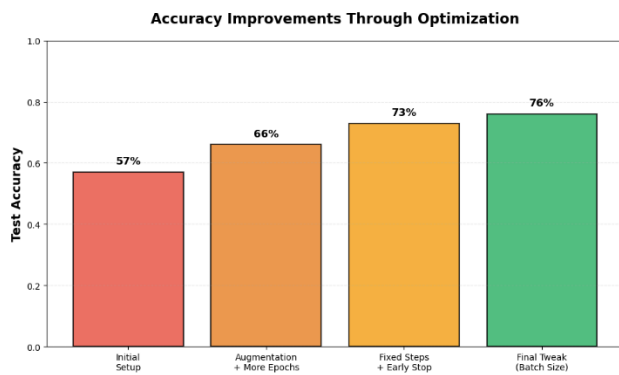


Fig. 2. Accuracy improvements across optimisation stages” in the figures list.

Grad-CAM / explainability paragraph: As a qualitative sanity check on model behaviour, class activation visualisation using Grad-CAM was applied to selected test radiographs. These heatmaps typically highlight lung regions and mediastinal areas that are relevant for the predicted pathologies, providing an intuitive indication of where the network is focusing and helping to assess whether decisions are driven by plausible anatomical patterns rather than spurious artefacts.



Fig. 3 Original X-ray plus several Grad-CAM heatmaps

VII. CONCLUSIONS

In this project, we implemented and evaluated a DenseNet-121-based model for multi-label chest X-ray classification on a reduced subset of the NIH ChestX-ray14 dataset. By combining transfer learning, class-weighted binary cross-entropy, data augmentation, and early stopping, we achieved a test binary accuracy of approximately 76%, a macro AUC of about 0.60, and an optimised macro F1-score of roughly 0.29. These results demonstrate that even under constrained computational resources and limited training data, a carefully tuned deep learning model can capture clinically relevant patterns in chest X-rays, while still falling short of the performance reported in large-scale studies that train on the full dataset. This review has surveyed current deep learning approaches for CXR analysis and discussed their potential to reshape image-based diagnosis. Although performance in research datasets is often strong, several technical and practical barriers still need to be overcome before such systems can be fully embedded in day-to-day clinical use. Addressing these issues in a structured way will be essential if DL is to support radiologists and improve patient outcomes rather than remain a purely experimental tool.

Strengths and limitations of current DL models: Many published CXR studies report high accuracy when deep-learning models are trained and evaluated on carefully selected datasets that emphasise a small set of relatively common diseases. Techniques such as transfer learning and data augmentation have helped boost accuracy in these controlled settings, but real-world CXRs often contain several overlapping abnormalities, and many models are still designed around single-label predictions. As a result, some systems may overlook co-existing diseases or provide only one label per study, which can reduce clinical usefulness in everyday practice where mixed pathology is frequent.

Future directions for model design: one path forward is to exploit the full probability outputs of multi-task or hybrid DL architectures so they can represent multiple simultaneous pathologies more explicitly. Extending analysis beyond the lungs to structures in the neck, mediastinum, and upper abdomen would also allow models to confirm truly normal CXRs or highlight incidental but important findings that might divert a reader’s attention. In addition, combining classification with localisation, segmentation, and quantitative measurements—for example, cardiothoracic ratio, nodule number and size, or pneumothorax volume—could provide clinicians with richer, structured information for diagnosis and treatment planning.

Overcoming the data bottleneck: progress is still constrained by the lack of large, strongly labelled, high-quality public CXR datasets that cover a broad spectrum of disease and imaging conditions. Curating such data requires expert readers, consistent labelling standards, and careful removal of artefacts, all of which are

time-consuming and expensive in a speciality that already faces workforce shortages. To reduce this burden, researchers are exploring semi- or unsupervised strategies and generative models such as GANs and variational autoencoders to help with data curation, augmentation, and label refinement.

Validation and implementation: before deployment, DL systems need robust external validation across different hospitals, scanners, and patient populations to confirm that performance generalises beyond the development cohort. Studies have shown that the same multi-class CXR model can achieve slightly different AUROC values in emergency departments compared with controlled study environments, underlining the importance of context-specific testing. Standardised benchmarking datasets, reporting guidelines, and common evaluation metrics are required so that results from different groups can be compared fairly and linked to clinically meaningful outcomes such as diagnostic accuracy, workflow impact, and cost-effectiveness.

Transparency, explainability, and the radiologist's role: many CXR AI papers still provide limited detail on model architecture, training data, and failure cases, which makes independent replication and extension difficult. Existing explainability tools—such as attention or saliency maps—offer a visual indication of which regions influenced a prediction, but they often do not fully clarify why an error occurred or how the model will behave on unfamiliar data. In practice, DL is most likely to function as an assistive “second reader,” helping radiologists prioritise studies, highlight subtle findings, and automate repetitive tasks such as measurements or preliminary reporting, while clinicians retain responsibility for integrating AI output with clinical history and other tests.

ETHICAL CONSIDERATIONS

Ethical and governance issues are crucial when developing and deploying deep-learning systems for chest X-ray analysis, because they operate on sensitive patient imaging data and can directly influence clinical decisions. Because these models rely on medical images, developers must strictly follow privacy and data-protection rules, ensuring that scans are properly de-identified, securely stored, and accessed only for clearly defined clinical or research purposes. Careful dataset curation is also essential to reduce demographic and clinical bias. If certain age groups, sexes, or communities are under-represented in the training data, models may perform worse for those patients, reinforcing existing inequalities in access to accurate diagnosis and timely treatment. Transparency and explainability form another key pillar, since opaque “black-box” predictions can erode clinician trust and make it difficult to recognise systematic errors or hidden biases. Wherever possible, systems should provide human-interpretable information about how decisions are reached and include clear lines of accountability for errors. Deep learning tools for CXR

interpretation should therefore be implemented as decision-support system, with clear clinical governance so that radiologists remain unlimitedly responsible for final reports and treatment decisions with using AI output among many.

VIII. REFERENCES

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